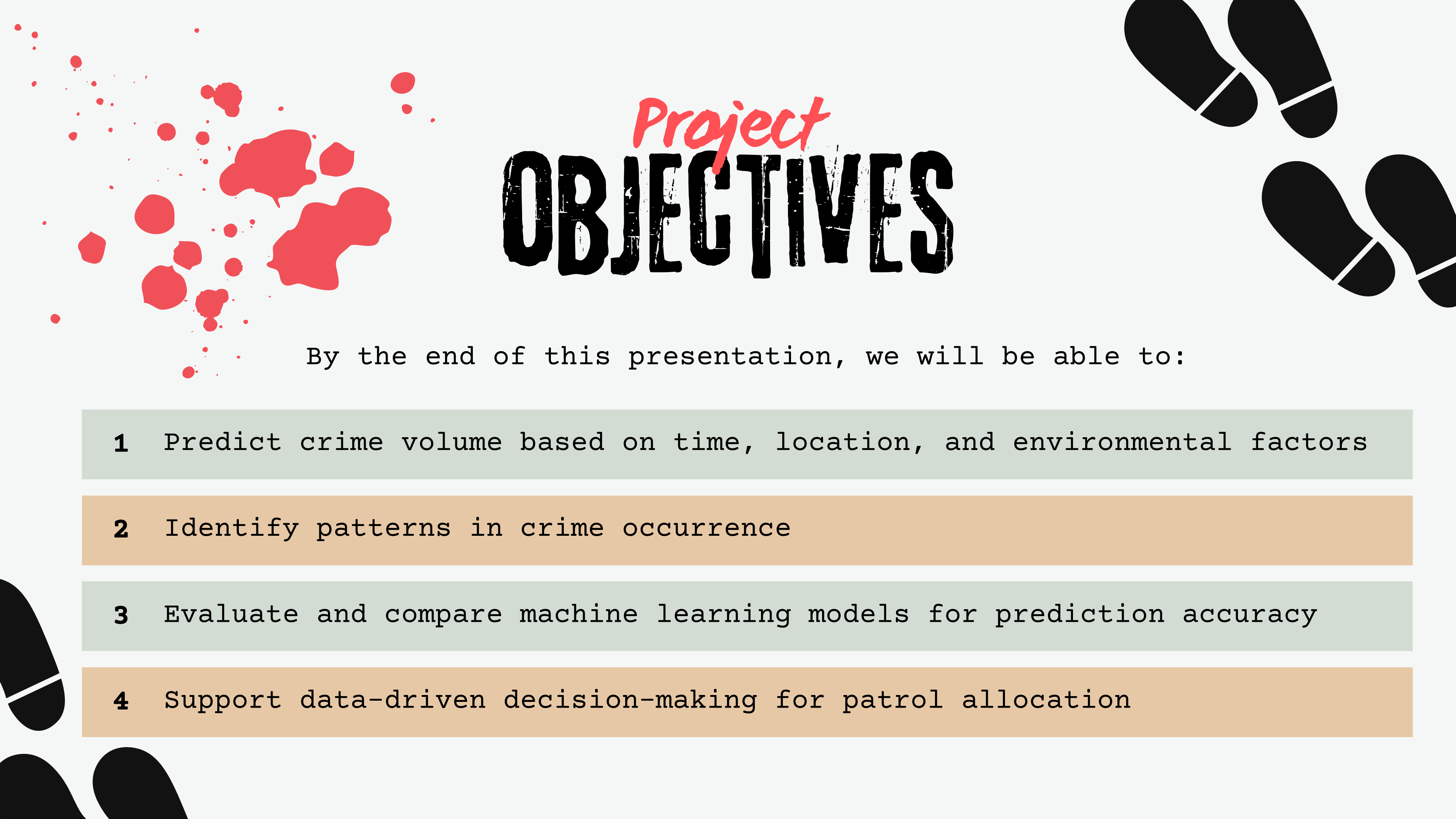


Presented by: Chanoudom Tann 6610755487

Apivit Denpruktham 6610755354

LAPD

CRIME PREDICTION

The slide features a white background with decorative elements. On the top left, there are several red splatters of varying sizes. On the top right, there are four black footprints arranged in a path. The title 'Project OBJECTIVES' is centered, with 'Project' in a red, handwritten-style font and 'OBJECTIVES' in a large, bold, black, distressed font.

Project OBJECTIVES

By the end of this presentation, we will be able to:

- 1 Predict crime volume based on time, location, and environmental factors
- 2 Identify patterns in crime occurrence
- 3 Evaluate and compare machine learning models for prediction accuracy
- 4 Support data-driven decision-making for patrol allocation

What are

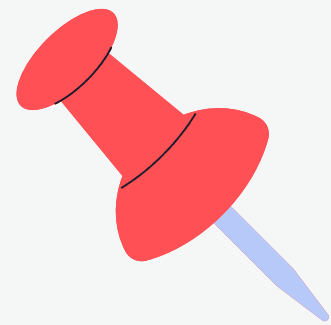
PROBLEM STATEMENTS

LAPD covers 21
geographic areas
across Los
Angeles

Current patrol
allocation is
reactive (respond
after crimes
occur)

Crime patterns
vary by time,
location, and
external factors

Need a way to
predict crime
volume in advance



Data

PREPARATION



Cleaned Dataset

Cleaned dataset to remove unreliable records

Data Filtering

- Filtered non-physical crimes (e.g., fraud, identity theft)

Time Extraction

- Extracted time-based features from date and time

Preparation

Prepared structured data for modeling

A Brief into **DATASET**

Crime Data(LAPD - data.gov)

- ~900,000+ crime records (2020-2023)
- Covers 21 areas in Los Angeles
- Includes: date, time, area, crime type

Weather Data (NOAA)

- Source: LAX weather station
- Features: temperature, rainfall, weather conditions

Final Dataset

- Merged crime + weather data by date

A Brief into DATASET

	DR_NO	Date Rptd	DATE OCC	TIME OCC	AREA	AREA NAME	Rpt Dist No	Part 1-2	Crm Cd	Crm Cd Desc ...	Premis Desc	Weapon Used Cd	Weapon Desc	Status	Status Desc	Crm Cd 1		LOCATION	LAT	LON	has_victim	
0	211507896	2021-04-11	2020-11-07	845	15	N Hollywood	1502	2	354	THEFT OF IDENTITY ...	SINGLE FAMILY DWELLING	NaN	NaN	IC	Invest Cont	354.0	7800	BEEMAN	AV	34.2124	-118.4092	1
1	201516622	2020-10-21	2020-10-18	1845	15	N Hollywood	1521	1	230	ASSAULT WITH DEADLY WEAPON, AGGRAVATED ASSAULT ...	SIDEWALK	200.0	KNIFE WITH BLADE 6INCHES OR LESS	IC	Invest Cont	230.0		ATOLL	AV	34.1993	-118.4203	1
2	240913563	2024-12-10	2020-10-30	1240	9	Van Nuys	933	2	354	THEFT OF IDENTITY ...	SINGLE FAMILY DWELLING	NaN	NaN	IC	Invest Cont	354.0	14600	SYLVAN	ST	34.1847	-118.4509	1
3	210704711	2020-12-24	2020-12-24	1310	7	Wilshire	782	1	331	THEFT FROM MOTOR VEHICLE - GRAND (\$950.01 AND ...	STREET	NaN	NaN	IC	Invest Cont	331.0	6000	COMEY	AV	34.0339	-118.3747	1
4	201418201	2020-10-03	2020-09-29	1830	14	Pacific	1454	1	420	THEFT FROM MOTOR VEHICLE - PETTY (\$950 & UNDER) ...	ALLEY	NaN	NaN	IC	Invest Cont	420.0	4700	LA VILLA MARINA	33.9813	-118.4350	1	

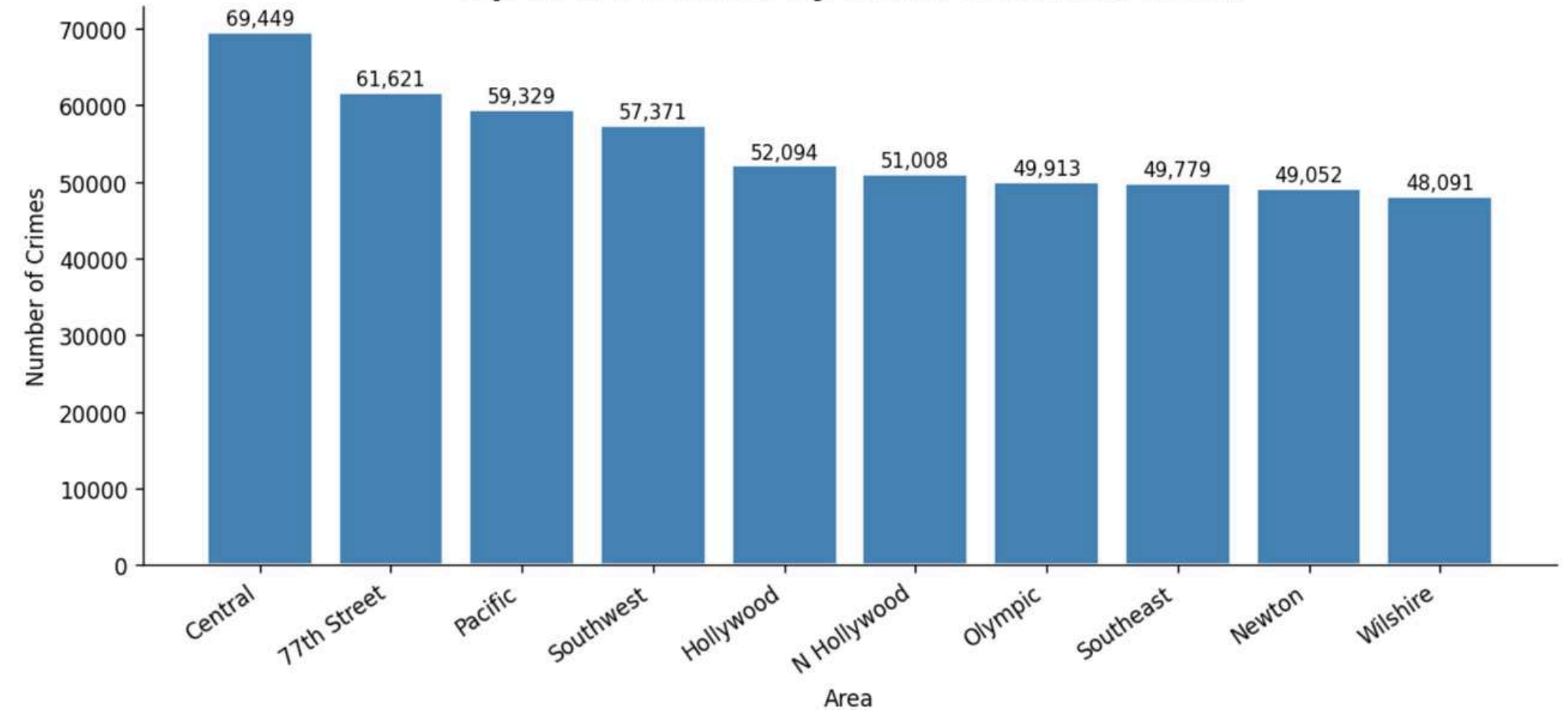
19 column

Key EDA INSIGHT

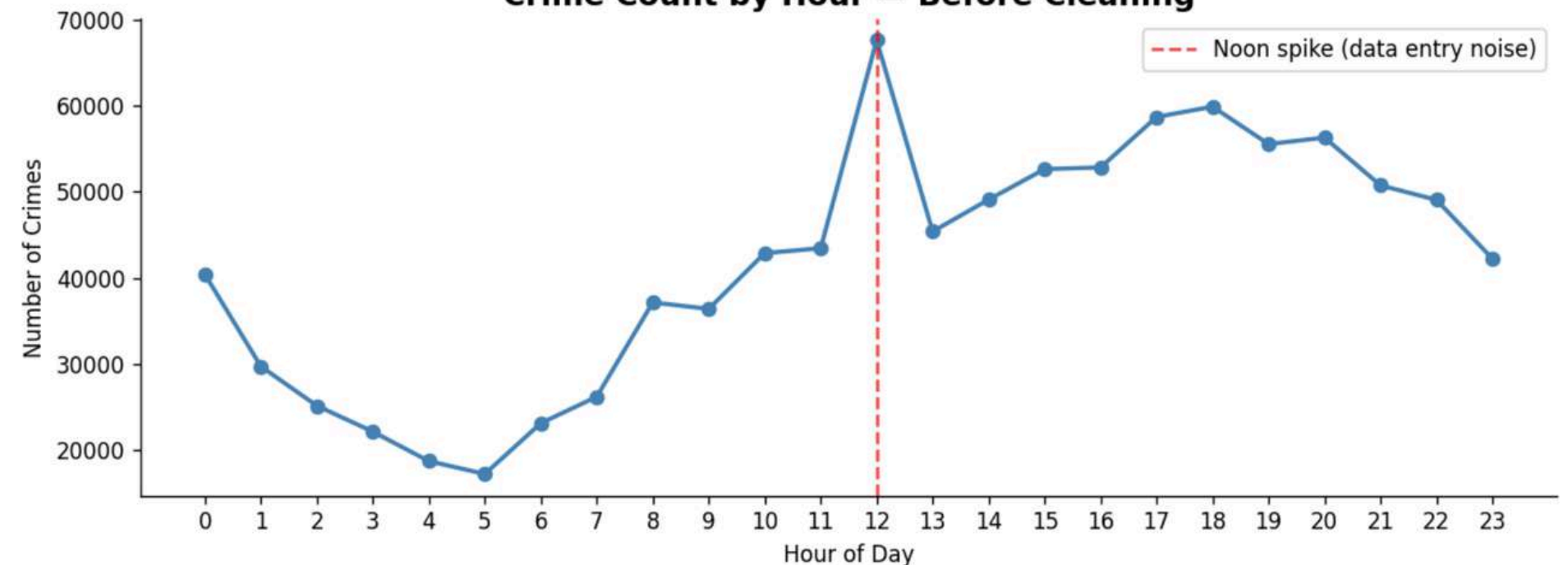
Content:

- Crime distribution varies significantly across areas
- Strong time-based patterns observed
- Certain hours consistently show higher crime levels
- Data anomalies were identified and removed

Top 10 LAPD Areas by Crime Count (All Years)

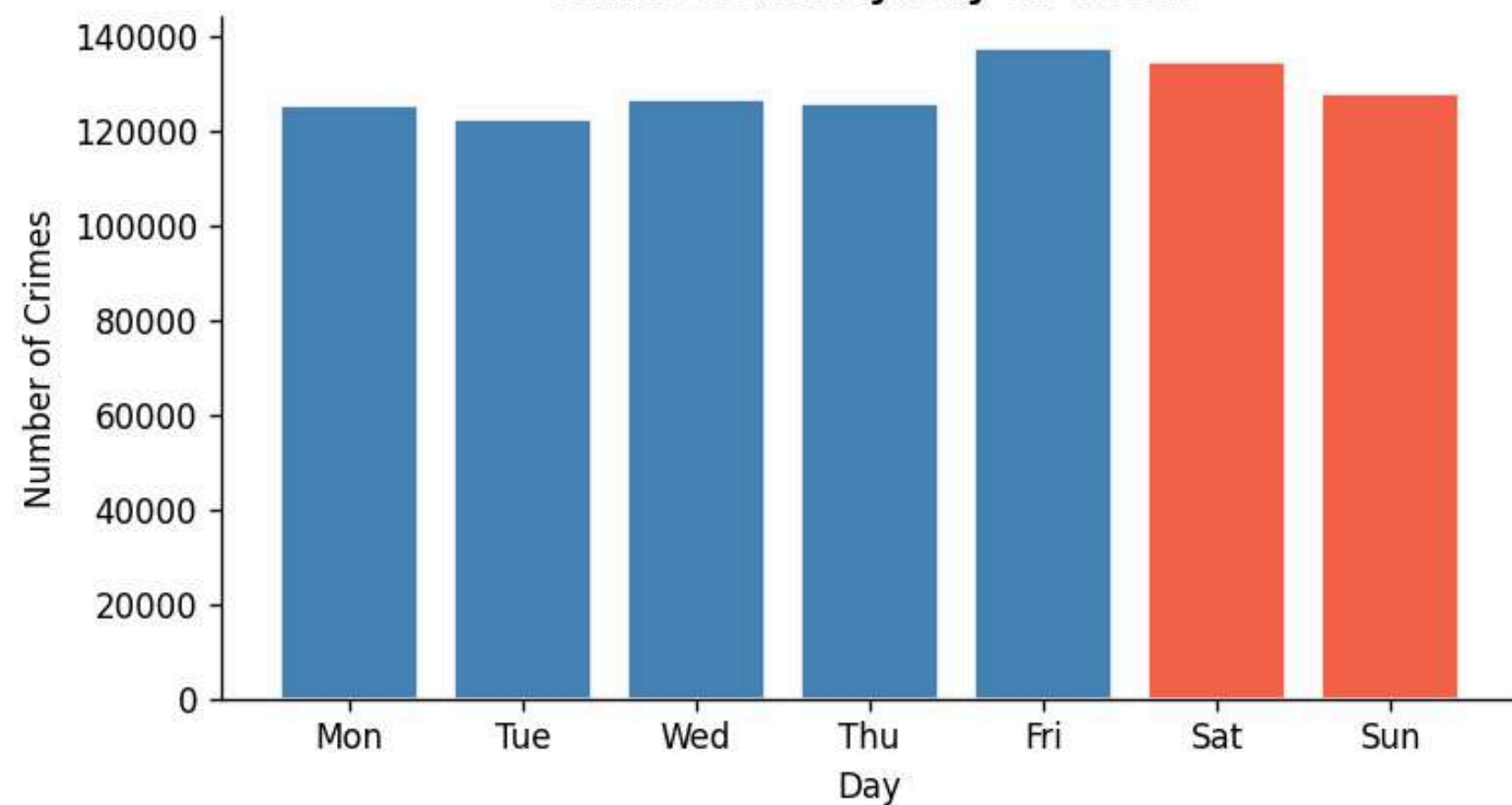


Crime Count by Hour — Before Cleaning

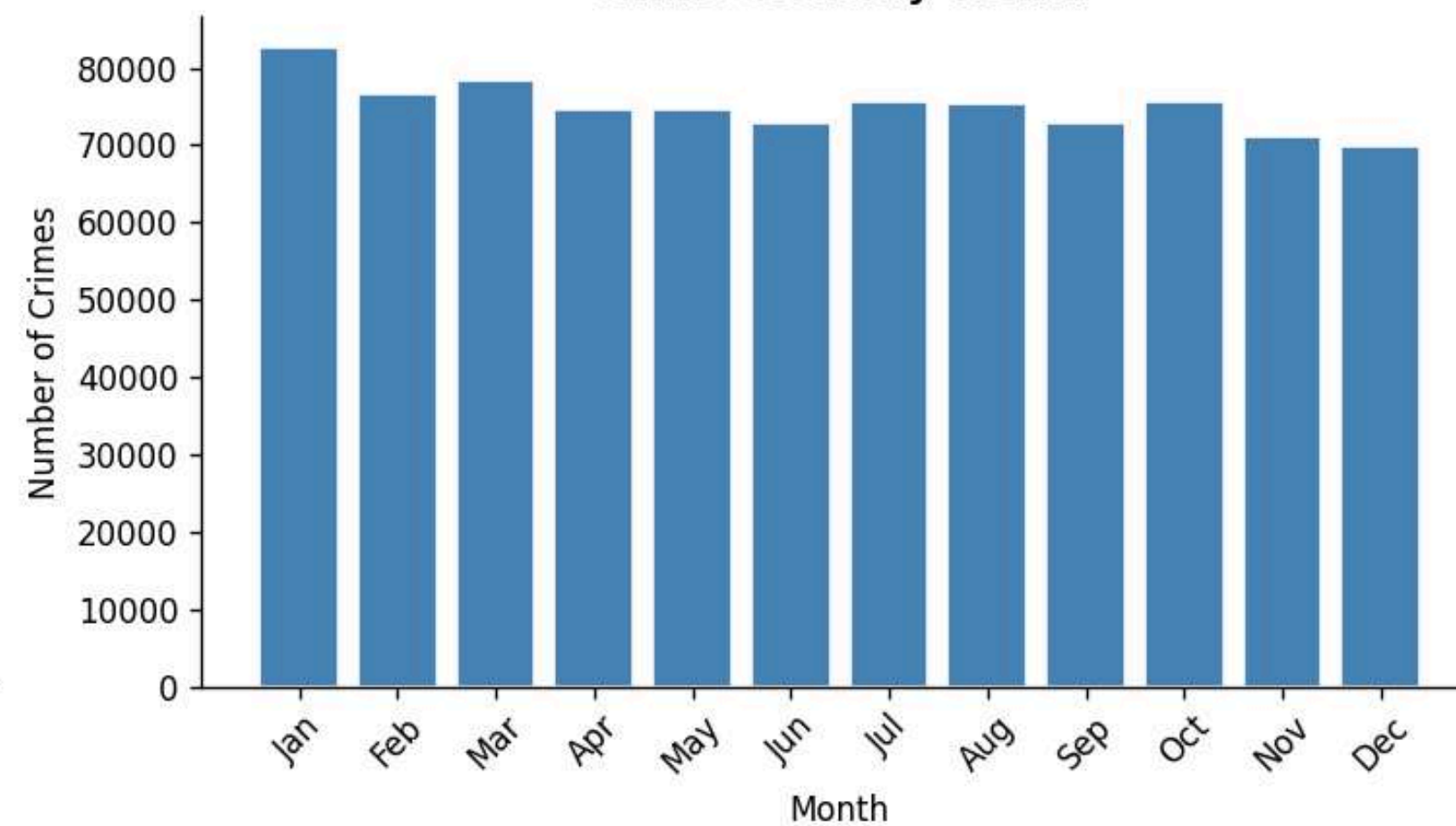


Key EDA INSIGHT

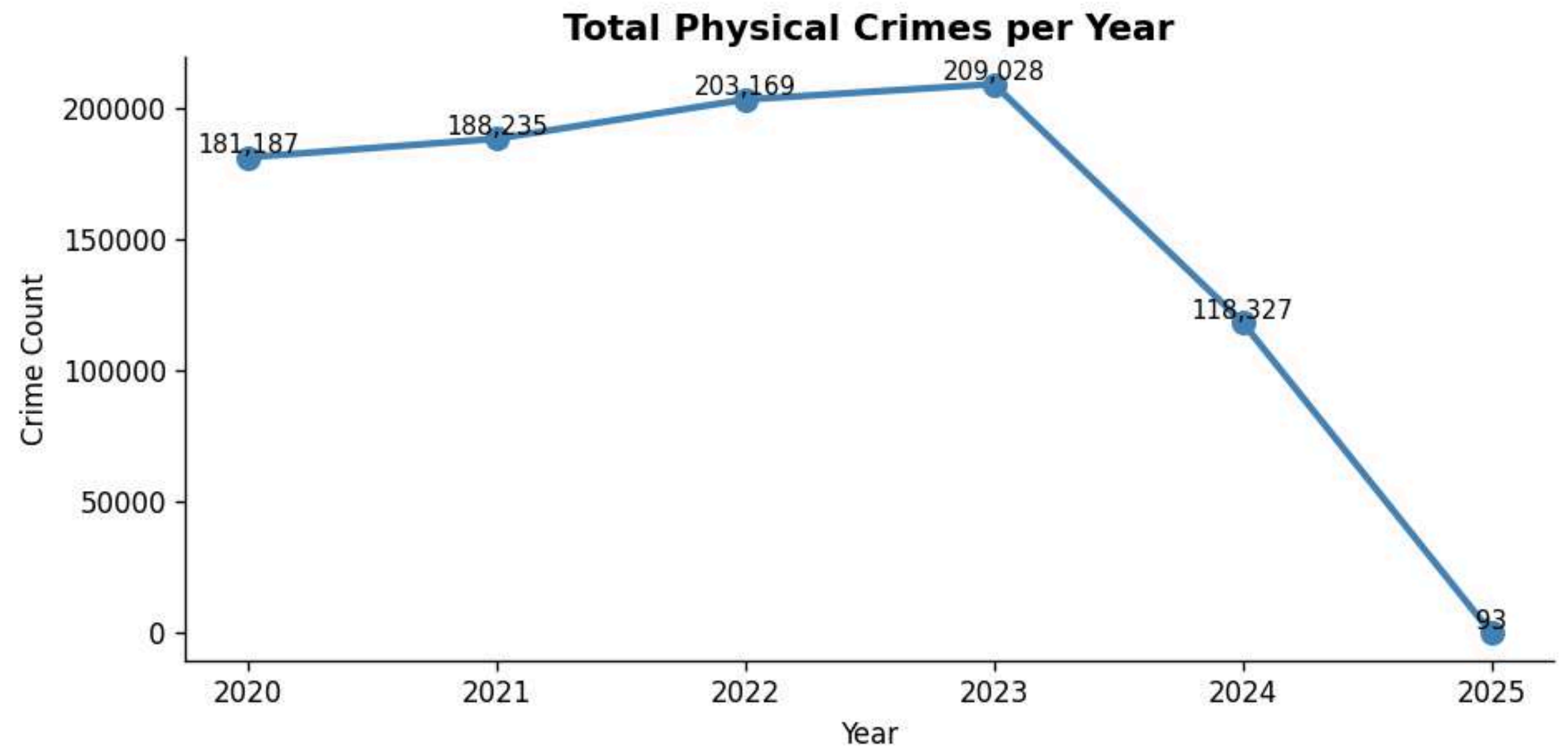
Crime Count by Day of Week



Crime Count by Month



Key EDA INSIGHT



Note: 2024–2025 are incomplete years – lower count does not mean lower crime

Feature ENGINEERING

Time & Calendar Features

- Hour grouped into 4 time buckets(Night, Morning, Afternoon, Evening)
- Day of week & month extracted
- Weekend indicator (0/1)
- Captures temporal crime patterns

Location Feature

- Area encoded using mean crime level
- (target encoding)
- Represents historical crime intensity
- Helps model learn spatial differences

Weather Features

- Temperature (TMAX, TMIN)
- Rainfall (PRCP)
- Weather indicators (rainy, hot)
- Holiday indicator added



Target VARIABLE



- We want to predict how many crimes will happen
- Data is grouped by:
 - Area
 - Day
 - Time period (morning, afternoon, etc.)
- Each row in our data means:
 - => number of crimes in one area at one time

Model Selection for cv

Evaluation metric:
Mean Absolute Error (MAE)



**Linear
Regression**

MAE = 2.13

Target: Predict crime count per area
and time slot

- Used Cross-Validation (3 folds)
- Data split into 3 parts
- Model trained and tested 3 times
- Results averaged for reliability
- Reduces overfitting



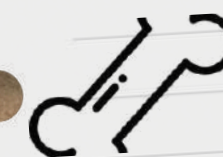
**Random
Forest**

MAE = 2.17



**Gradient
Boasting**

MAE = 1.93



XGBoost

MAE = 1.91

```

from sklearn.ensemble import RandomForestRegressor, GradientBoostingRegressor
from sklearn.linear_model import LinearRegression
from sklearn.model_selection import cross_val_score
from xgboost import XGBRegressor
import numpy as np

X = target[['area_code', 'year', 'month', 'day_of_week', 'hour']]
y = target['crime_count']

models = {
    'Linear Regression': LinearRegression(),
    'Gradient Boosting': GradientBoostingRegressor(n_estimators=100, random_state=42),
    'Random Forest': RandomForestRegressor(n_estimators=100, random_state=42, n_jobs=-1),
    'XGBoost': XGBRegressor(n_estimators=100, random_state=42),
}

hourly_results = {}
for name, model in models.items():
    scores = cross_val_score(model, X, y, cv=3, scoring='neg_mean_absolute_error')
    mae = -scores.mean()
    hourly_results[name] = mae
    print(f'{name}: MAE = {mae:.2f}')

```

Linear Regression: MAE = 2.13
 Gradient Boosting: MAE = 1.93
 Random Forest: MAE = 2.17
 XGBoost: MAE = 1.91

```

from xgboost import XGBRegressor

xgb = XGBRegressor(n_estimators=100, random_state=42)
scores = cross_val_score(xgb, X, y, cv=3, scoring='neg_mean_absolute_error')
print(f'XGBoost MAE = {-scores.mean():.2f}')

```

Selected Model: **XGBoost**

- Lowest MAE
- Captures complex (non-linear) patterns
- Effective for structured data
- Built-in regularization reduces overfitting
- Selected based on cross-validation results



CROSS-VALIDATION MODEL COMPARISON (3-FOLD CV)

```
X2 = target2[['area_code', 'year', 'month', 'day_of_week', 'hour_bucket']]
y2 = target2['crime_count']

for name, model in models.items():
    scores = cross_val_score(model, X2, y2, cv=3, scoring='neg_mean_absolute_error')
    print(f'{name}: MAE = {-scores.mean():.2f}')

# XGBoost too
xgb2 = XGBRegressor(n_estimators=100, random_state=42)
scores = cross_val_score(xgb2, X2, y2, cv=3, scoring='neg_mean_absolute_error')
print(f'XGBoost: MAE = {-scores.mean():.2f}')
```

Purpose of this experiment

- Tested time buckets instead of exact hour
- Compared model performance using 3-fold CV
- Result was worse than exact-hour model
- Shows that feature design strongly affects model accuracy

Better Result of MAE



LINEAR REGRESSION: 8.38



GRADIENT BOOSTING: 7.23



RANDOM FOREST: 7.86



XGBOOST: 7.66

Improved Model With FEATURE ENGINEERING

Replaced area_code with area mean crime
(target encoding)

Added weekend indicator

Used time-based features (month, day,
hour bucket)



```
In [29]: # 1. Target encoding: replace area_code with area's mean crime count
area_mean = target2.groupby('AREA NAME')['crime_count'].mean()
target2['area_mean_crime'] = target2['AREA NAME'].map(area_mean)

# 2. Add is_weekend
target2['is_weekend'] = (target2['day_of_week'] >= 5).astype(int)

X3 = target2[['area_mean_crime', 'year', 'month', 'day_of_week', 'hour_bucket', 'is_weekend']]
y3 = target2['crime_count']

for name, model in models.items():
    scores = cross_val_score(model, X3, y3, cv=3, scoring='neg_mean_absolute_error')
    print(f'{name}: MAE = {-scores.mean():.2f} ({-scores.mean()/target2["crime_count"].mean()*100:.1f}%)')

xgb3 = XGBRegressor(n_estimators=100, random_state=42)
scores = cross_val_score(xgb3, X3, y3, cv=3, scoring='neg_mean_absolute_error')
print(f'XGBoost: MAE = {-scores.mean():.2f} ({-scores.mean()/target2["crime_count"].mean()*100:.1f}%)')
```

```
Linear Regression: MAE = 7.45 (29.3%)
Gradient Boosting: MAE = 5.83 (22.9%)
Random Forest: MAE = 6.16 (24.2%)
XGBoost: MAE = 5.54 (21.8%)
```

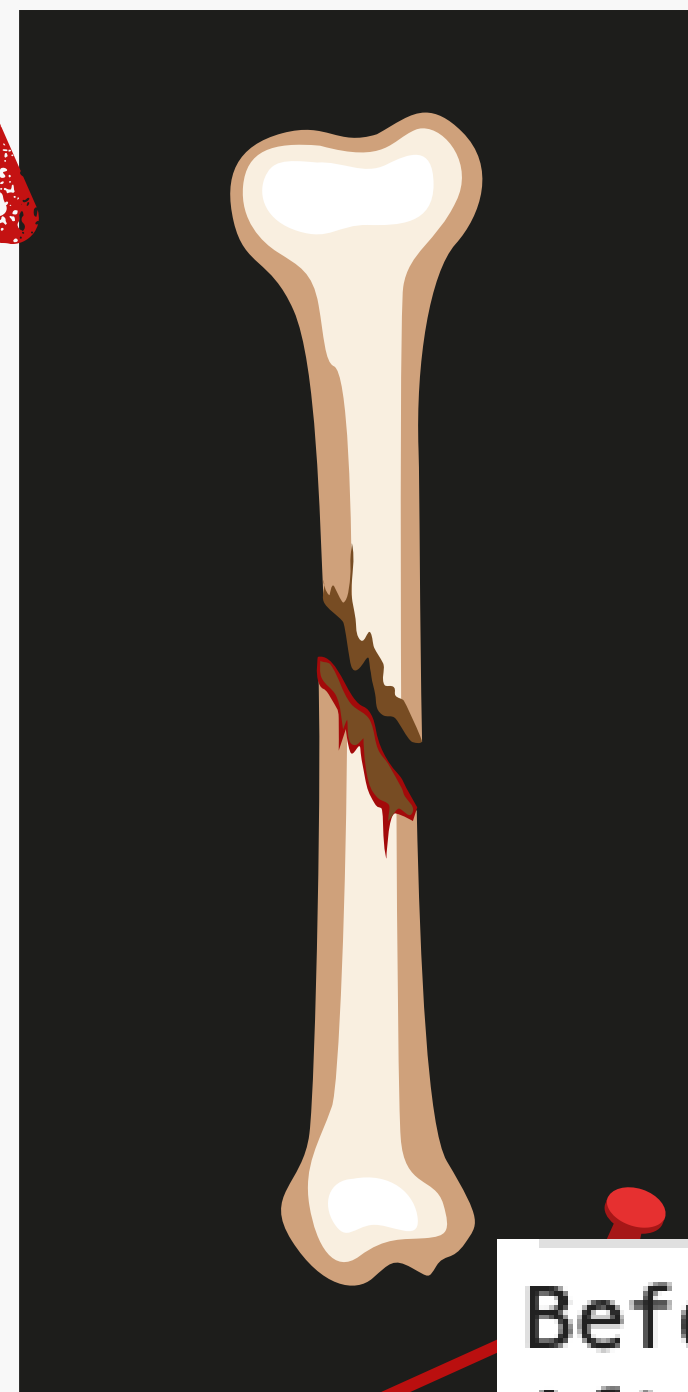
INTEGRATING WEATHER DATA

Data Integration

- Combined crime data with daily weather data
- Merged based on date

Weather Features Added

- Temperature (TMAX, TMIN)
- Rainfall (PRCP)
- Weather conditions (rainy, hot, etc.)



Continue

Data Quality Check

- Total records unchanged: 900,039
- No missing weather values after merge

Before merge: 900039
After merge: 900039
Weather nulls after merge: 0

FINAL DATASET CONSTRUCTION

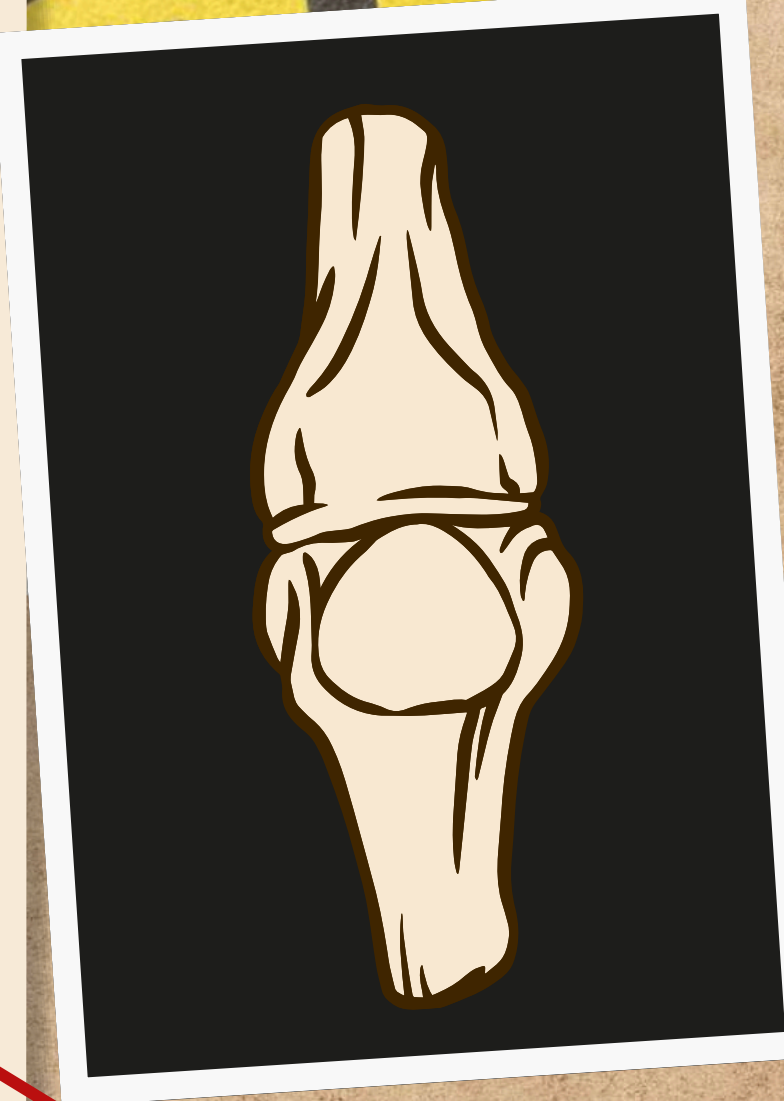
GROUPED DATA BY:

- AREA
- DATE (YEAR, MONTH, DAY)
- TIME SLOT (HOUR BUCKET)

CLUE # 4

FEATURES COMBINED

- TIME FEATURES (HOUR, DAY, MONTH)
- LOCATION (AREA)
- WEATHER (TEMPERATURE, RAINFALL, CONDITIONS)



```
# Add hour_bucket to crime_weather
crime_weather['hour_bucket'] = crime_weather['hour'].apply(hour_to_bucket)

# Aggregate: crime count + weather per area per time slot
target3 = crime_weather.groupby(['AREA NAME', 'year', 'month', 'day_of_week', 'hour_bucket']).agg(
    crime_count=('Crm Cd', 'count'),
    TMAX=('TMAX', 'mean'),
    TMIN=('TMIN', 'mean'),
    PRCP=('PRCP', 'mean'),
    is_rainy=('is_rainy', 'max'),
    is_hot=('is_hot', 'max'),
    WT01=('WT01', 'max'),
    WT03=('WT03', 'max'),
    WT08=('WT08', 'max'),
).reset_index()

# Target encoding – use training years only to avoid leakage from incomplete 2024–2025
area_mean3 = target3[target3['year'] <= 2022].groupby('AREA NAME')['crime_count'].mean()
target3['area_mean_crime'] = target3['AREA NAME'].map(area_mean3)
target3['is_weekend'] = (target3['day_of_week'] >= 5).astype(int)

print(target3.shape)
target3.head()
```

(35345, 16)

(35345, 16)

	AREA NAME	year	month	day_of_week	hour_bucket	crime_count	TMAX	TMIN	PRCP	is_rainy	is_hot	WT01	WT03	WT08	area_mean_crime	is_weekend
0	77th Street	2020	1	0	0	20	67.800000	51.400000	0.008000	1	0	1	0	1	36.059524	0
1	77th Street	2020	1	0	1	29	65.413793	51.241379	0.006207	1	0	1	0	1	36.059524	0
2	77th Street	2020	1	0	2	58	68.000000	50.844828	0.005862	1	0	1	0	1	36.059524	0
3	77th Street	2020	1	0	3	65	70.092308	49.984615	0.004000	1	0	1	0	1	36.059524	0
4	77th Street	2020	1	1	0	28	69.214286	53.642857	0.007500	1	0	0	0	1	36.059524	0

FINAL MODEL TRAINING WITH WEATHER FEATURE

Model: XGBoost

Features:

- Time (month, day, hour bucket)
- Location (area mean encoding)
- Weather (temperature, rainfall, conditions)

Performance (Cross-Validation)

MAE = 5.51

ERROR $\approx$ 21.7

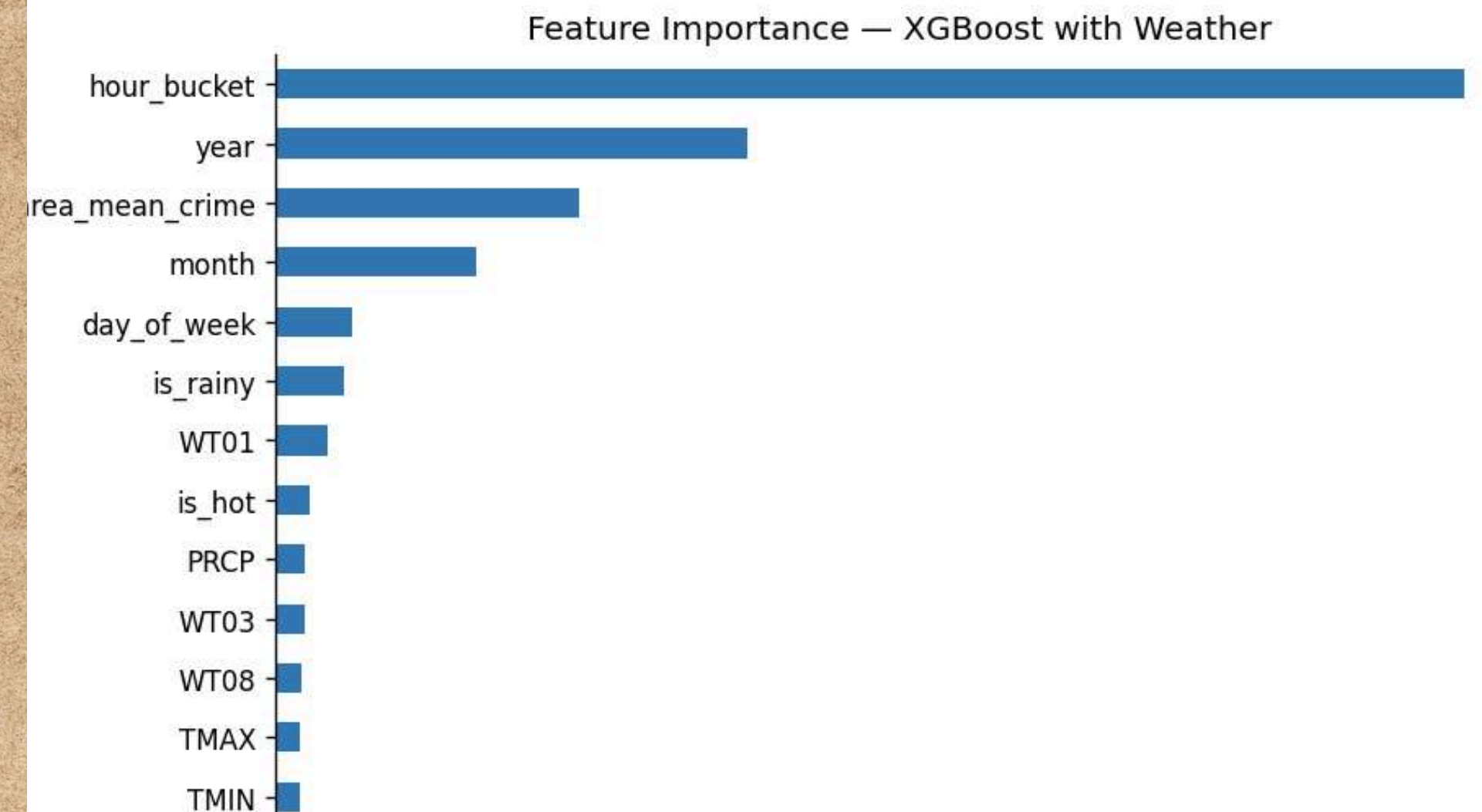
FEATURE IMPORTANCE ANALYSIS

Top Influencing Factors

- Hour bucket (time of day) → most important
- Year → captures long-term trends
- Area (location) → crime varies by region
- Month → seasonal patterns

Weather Impact

- Rain and temperature have smaller but noticeable effects
- Weather contributes additional context to predictions



WHY WE USED A TIME-BASED SPLIT (NOT RANDOM CV)

-
-
-
- **Crime is time-series data. Random folds let the model peek at the future — that's data leakage.**
-
-
-

Crime = time-series data

- Random CV trains on 2023, tests on 2020 → data leakage
- Time-split trains on 2020–22, tests on 2023 → realistic

Proof

- XGBoost: best on CV (21.8%) → 4th on time-split (22.7%)
- Gradient Boosting: 21.8% on time-split ✓
- Final
- Reported accuracy: 21.8% MAE (GB, time-split)
- Matches real-world deployment

TIME-BASED MODEL COMPARISON

Evaluation Method

Train: 2020–2022

Test: 2023

Metric: Mean Absolute Error (MAE)

LINEAR REGRESSION: MAE =

2.23(43.4% OF MEAN)

GRADIENT BOOSTING: MAE = 1.96

(38.1% OF MEAN)

RANDOM FOREST: MAE = 2.20

(42,8% OF MEAN)

XGBOOST: MAE = 1.97

(38.3% OF MEAN)

```
# Time-based train/test on hourly target (train 2020-2022, test 2023)
# Year is dropped - tree models can't extrapolate to an unseen year
from sklearn.metrics import mean_absolute_error

features_tt = ['area_code', 'month', 'day_of_week', 'hour']
train_h = target[target['year'] <= 2022]
test_h = target[target['year'] == 2023]
test_h_mean = test_h['crime_count'].mean()

models_tt = {
    'Linear Regression': LinearRegression(),
    'Gradient Boosting': GradientBoostingRegressor(n_estimators=100, random_state=42),
    'Random Forest': RandomForestRegressor(n_estimators=100, random_state=42, n_jobs=-1),
    'XGBoost': XGBRegressor(n_estimators=100, random_state=42),
}

hourly_tt = {}
for name, model in models_tt.items():
    model.fit(train_h[features_tt], train_h['crime_count'])
    preds = model.predict(test_h[features_tt])
    mae = mean_absolute_error(test_h['crime_count'], preds)
    hourly_tt[name] = mae
    print(f'{name}: MAE = {mae:.2f} ({mae/test_h_mean*100:.1f}% of 2023 mean)')

print(f'\n2023 mean crime count: {test_h_mean:.2f}')
```

Python

```
Linear Regression: MAE = 2.23 (43.4% of 2023 mean)
Gradient Boosting: MAE = 1.96 (38.1% of 2023 mean)
Random Forest: MAE = 2.20 (42.8% of 2023 mean)
XGBoost: MAE = 1.97 (38.3% of 2023 mean)
```

```
2023 mean crime count: 5.14
```

```
# XGBoost with weather features (cross-validation)
X_w = target3[['area_mean_crime', 'year', 'month', 'day_of_week', 'hour_bucket', 'is_weekend',
               'TMAX', 'TMIN', 'PRCP', 'is_rainy', 'is_hot', 'WT01', 'WT03', 'WT08']]
y_w = target3['crime_count']

xgb4 = XGBRegressor(n_estimators=200, max_depth=5, learning_rate=0.1, random_state=42)
scores = cross_val_score(xgb4, X_w, y_w, cv=3, scoring='neg_mean_absolute_error')
print(f'XGBoost with weather: MAE = {-scores.mean():.2f} ({-scores.mean()/target3["crime_count"].mean()*100:.1f}%)')
```

XGBoost with weather: MAE = 5.51 (21.7%)

TIME BASED EVALUATION SUMMARY

Evaluation Setup

- Train: 2020–2022
- Test: 2023
- Year feature removed

Final Performance

- MAE = 6.78
- Average crime count ≈ 29.6
- Error $\approx 22.9\%$

HOLIDAY FEATURE INTEGRATION

Final Dataset Enhancement

- Added US federal holiday feature
- Rebuilt aggregated dataset by area, date, and time bucket
- Combined crime, weather, weekend, and holiday features
- Final dataset: 35,345 rows and 17 columns

Purpose:

To help the model capture changes in crime patterns during holidays and weekends.



IMPACT OF HOLIDAY FEATURE ON MODEL PERFORMANCE

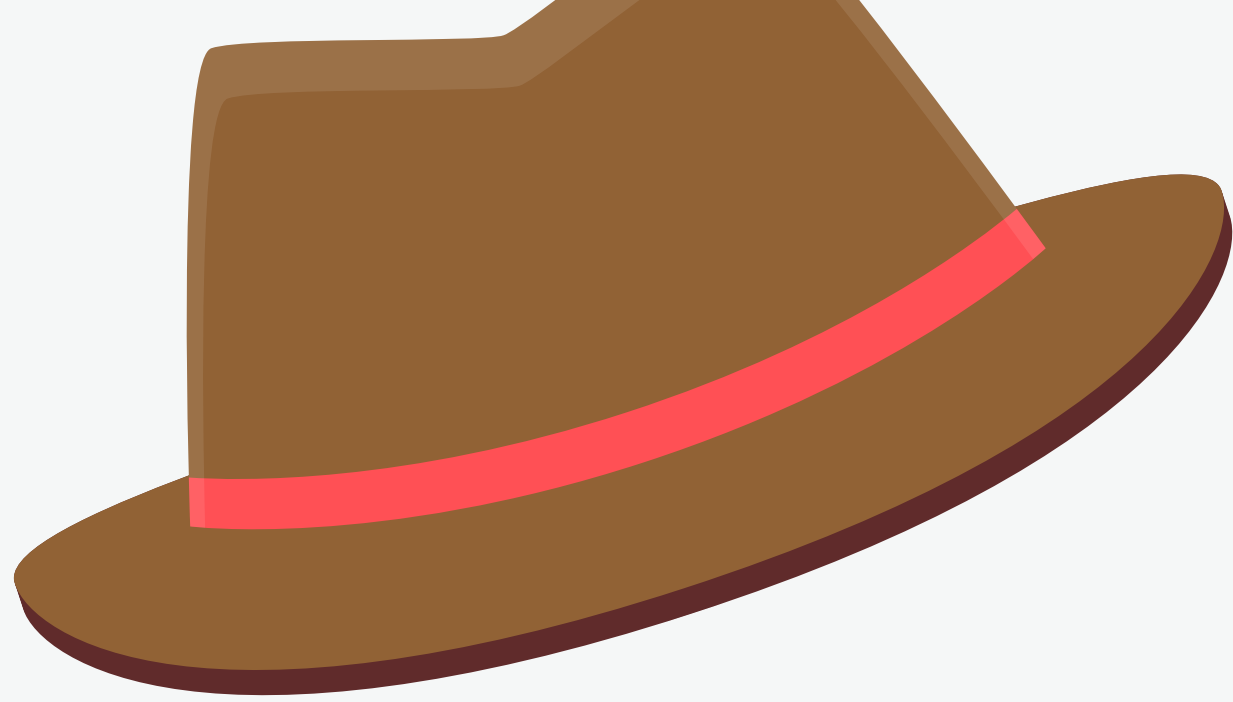


Evaluation Setup

- Time-based split (2020–2022 → 2023)
- Metric: MAE

Results

- Without holiday: 22.9%
- With holiday: 22.7%



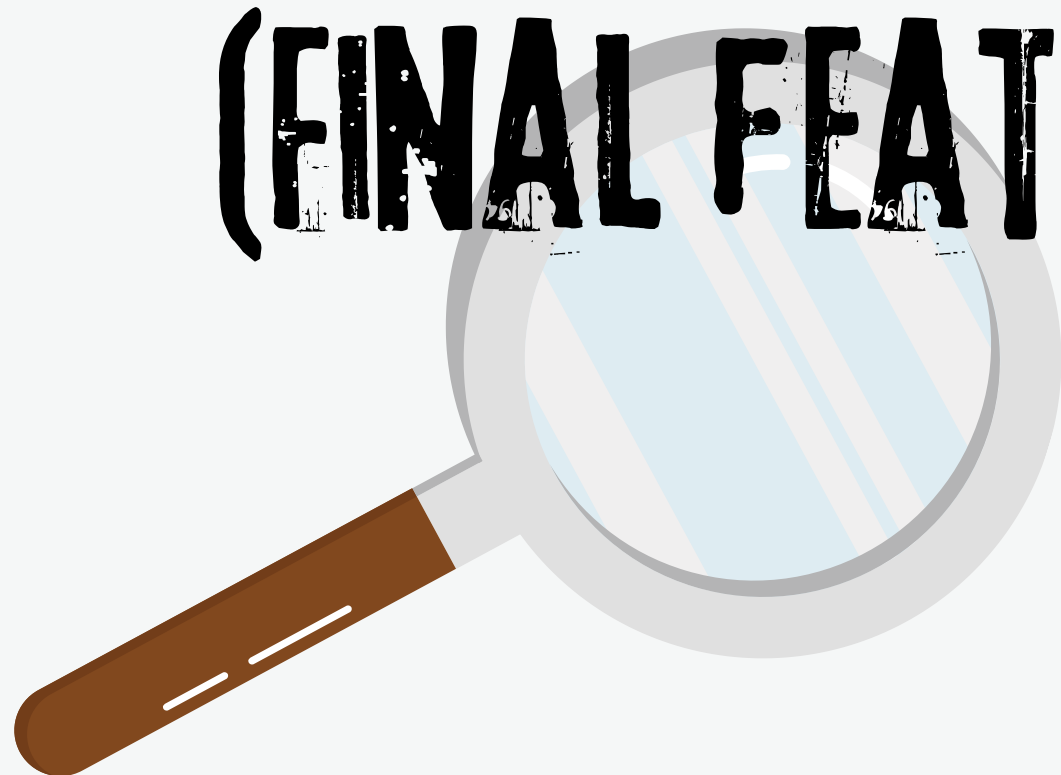
Evaluation Setup

- Train: 2020–2022
- Test: 2023
- Metric: MAE

Time-Based model

COMPARISON

(FINAL FEATURE)

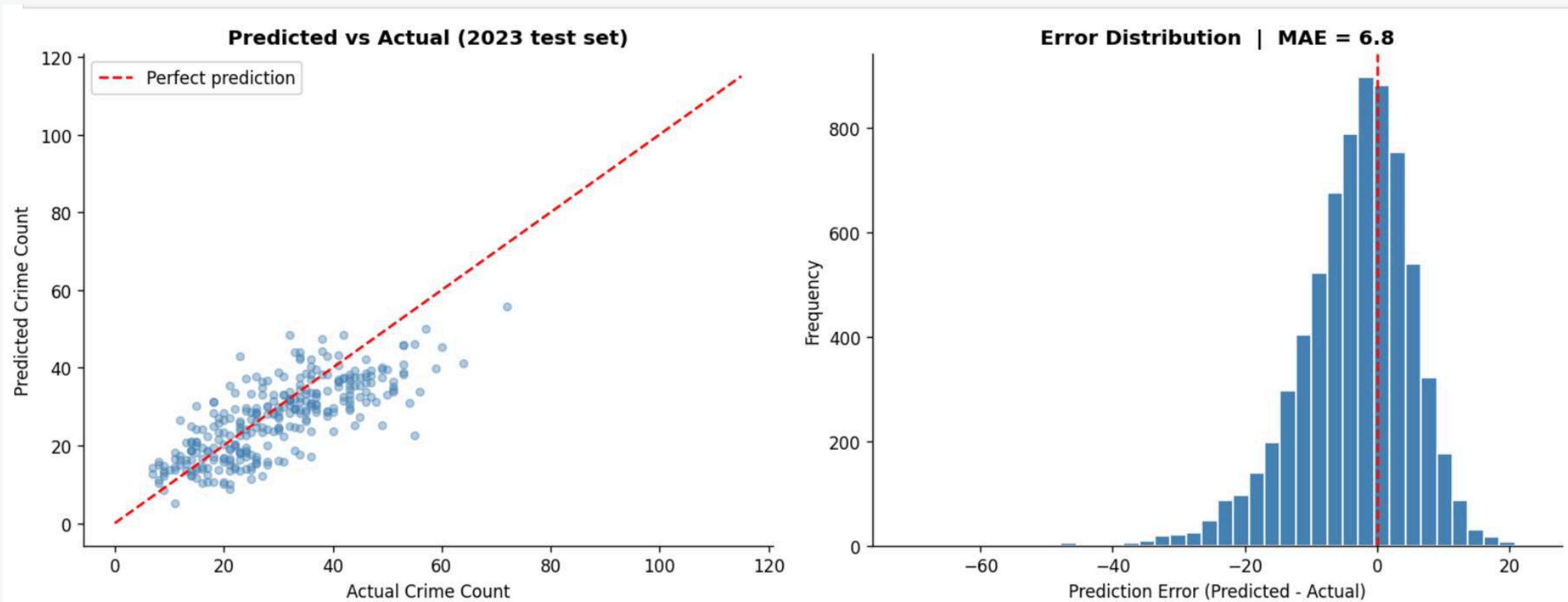


Linear Regression: MAE = 6.89 (23.2% of 2023 mean)
Gradient Boosting: MAE = 6.47 (21.8% of 2023 mean)
Random Forest: MAE = 6.58 (22.2% of 2023 mean)
XGBoost: MAE = 6.73 (22.7% of 2023 mean)

2023 mean crime count: 29.62



Final Model EVALUATION



CRIME PREDICTION DEMO

Area: 77th Street

Time: Friday Afternoon (12-17)

Weather:

Temperature: 85°F / 65°F

Rainfall: 0 inches

Holiday: No



Prediction

**Predicted crime count: ~44 incidents
combine with our 6.47 from mae so the actual
crime is likely in the range of 38-51**

SUMMARY

- Predicted crime count using time, location, and weather
- Cleaned data and removed noise (non-physical crimes)
- Engineered key features (time, location, weather, holiday)
- Compared models → selected XGBoost
- Final result: MAE ≈ 6.78 (~22.9% error)
- Key insight: Time and location are most important
- Use case: Supports patrol planning

thank you for
LISTENING

